LOADSHEDDING IMPACT ANALYSIS

Findings, Insights, Recommendations & Data Cleaning Methodology

Small-Business Sales Impact Study | Limpopo & Gauteng, South Africa

Analysis Period: 3 January 2023 – 16 February 2023

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Power BI Data Analytics Portfolio Project

Executive Summary

This report analyses the relationship between South African loadshedding (rolling electricity outages) and daily sales performance across four small businesses in Limpopo and Gauteng. Using a 130-row cleaned dataset spanning six weeks, the analysis quantifies revenue loss by loadshedding severity, tests the protective effect of backup generators, and identifies which business types are most exposed.

-66%	13x	R1.0m	97.7%
Sales drop at Severe stage (5-6) vs. no loadshedding	Sales advantage for generator-backed business at Stage 3+	Total sales analysed across 4 businesses	Data retained after cleaning (130 of 133 rows)

The core finding is that loadshedding's revenue impact is not flat — it compounds with severity, and it falls hardest on small, informal, and service-based businesses that lack backup power. Businesses with generators recover a large share of lost trade at high loadshedding stages, while service businesses with no non-electric fallback (such as a hair salon) can lose trading days entirely.

1. Data Cleaning Methodology

The raw export contained 133 rows pulled from four businesses' point-of-sale logs. Before any analysis, the dataset was audited and cleaned in Power Query (M), with every change logged for auditability rather than applied silently.

133	14	3	130
Raw rows in source export	Distinct data quality issues identified	Rows excluded (outlier, duplicate, invalid)	Clean rows carried into analysis

1.1 Data Quality Log

Each issue was categorised by type, field, and the action taken. A summary of all 14 issues:

#	Issue	Field	Action Taken	Status
1	3 inconsistent date formats	Date	Parsed & normalised to a single date type	Fixed
2	5 spelling/case variants of business names	Business_Name	Mapped to canonical names	Fixed
3	Inconsistent business type casing/truncation	Business_Type	Title-cased; "Spaza" expanded	Fixed
4	Province typos ("limpopo", "GP")	Province	Standardised to full names	Fixed
5	All-caps city entries	City	Title-cased	Fixed
6	5 encodings of a binary field	Generator_YN	Standardised to Yes/No	Fixed
7	Repeated CSV header row mid-file	All columns	Removed	Removed
8	Outlier: R999,999 (normal range R900-R3,100)	Daily_Sales_ZAR	Excluded; logged separately	Excluded
9	Exact duplicate row	All columns	Second occurrence removed	Removed
10	Negative sales value (R-150, refund day)	Daily_Sales_ZAR	Excluded; logged separately	Excluded
11	1 missing sales value on a Stage 4 day	Daily_Sales_ZAR	Imputed with business median, flagged	Imputed
12	1 missing avg transaction value	Avg_Transaction_Value	Recalculated as Sales ÷ Transactions	Recalculated
13	Wrong province abbreviation	Province	Corrected to "Gauteng"	Fixed
14	Inconsistent closed-day encoding (0/blank/N-A)	Sales/Transactions	Standardised: zeros = valid closed day	Standardised

Full detail, including exact row references, is maintained in the companion workbook's Data Quality Log tab.

1.2 Rows Removed

Three rows were excluded from analysis. Each is preserved in a separate audit log rather than deleted outright:

Date	Business	Value	Issue
2023-01-16	Thabo's Spaza Shop	R999,999	Outlier — suspected data entry error
2023-01-24	Thabo's Spaza Shop	R2,300	Exact duplicate of an existing row
2023-02-07	Thabo's Spaza Shop	-R150	Negative sales — refund day, not valid trade

1.3 Power Query (M) Pipeline

The cleaning steps were implemented as a single, reproducible M script so the transformation can be re-run end-to-end on any future export rather than repeated as manual edits. The pipeline, in order:

- Load raw CSV and promote headers
- Remove the repeated header row
- Parse and normalise all date formats
- Standardise Business_Name, Business_Type, Province, City, Generator_YN
- Convert numeric columns to correct types (whole numbers vs. decimals)
- Remove the outlier, duplicate, and negative-sales rows
- Add derived columns: Day_of_Week, Week_Number, Month
- Add LS_Severity band (None / Low 1-2 / High 3-4 / Severe 5-6) from Loadshedding_Stage

Why this matters for the analysis

- Standardising business names/types prevented the same business from being split into 3-5 phantom entities in the model.
- Excluding (not silently averaging over) the R999,999 outlier and negative-sales row protects every downstream average from distortion.
- The LS_Severity band, engineered here, is the field used throughout Sections 3-4 to group findings by outage severity.

2. Key Findings

2.1 Sales fall as loadshedding stage rises — consistently, across every business

Average daily sales at Stage 0 (no loadshedding) were R10,808, falling to R3,668 at Stage 5-6 (severe) — a 66% decline. Every business shows a strong negative correlation between stage and sales (r = -0.86 to -0.97), so this is a universal pattern, not one outlier skewing the average.

2.2 The revenue hit compounds with severity

- Stage 1-2 (Low): sales down ~13% vs. baseline
- Stage 3-4 (High): sales down ~39%
- Stage 5-6 (Severe): sales down ~66%

Each additional stage costs roughly another 8-10 percentage points of revenue — the relationship is not a flat "loadshedding tax".

2.3 Generators substantially cushion the impact — but only at high stages

During Stage 3+ loadshedding, businesses with a generator averaged R17,140/day versus R1,268/day without — over 13x higher. Caveat: only the retail store (Pick n Pay) has a generator, so this is confounded with business size and should be read as directional, not a clean causal effect.

2.4 Staffing appears to shrink alongside severity

Staff on duty averages 5.6 at Stage 0, falling to 2.4 at Stage 6 — a 58% reduction. This likely compounds the sales decline: fewer staff during outages may mean reduced hours or sections closed, on top of the power loss itself.

2.5 Small, informal and service businesses are hit hardest

- Lebo's Hair Salon: -57% sales impact, 3 zero-sales days, no generator, only 3.6% of group revenue
- Pick n Pay Express: -36% impact (smallest decline), retains 81% of group revenue, has a generator

A hair salon cannot deliver its service at all without power, unlike a spaza shop that can still sell some stock in the dark.

2.6 Zero-sales days cluster at the most severe stages

All 3 zero-sales days in the dataset occurred at Lebo's Hair Salon, at Stage 4, 5 and 6 respectively (8-12 hours without power) — directly tied to "closed due to loadshedding" notes in the source data.

2.7 Friday is the strongest trading day; midweek is weakest

Friday averaged R9,304 vs. R6,904 on Wednesday. A fairly typical retail pattern, but it means a loadshedding schedule that disproportionately hits Fridays would do outsized damage to weekly revenue.

3. Insights — So What?

Loadshedding is a regressive shock, not a uniform one

- Larger, capital-backed businesses (generator, more staff, diversified stock) absorb outages far better than small or informal businesses, widening the gap between them over time.

A generator's value is highest exactly when it's needed most

- The sales uplift from having a generator only shows up at Stage 3+, meaning a generator that sits idle during Stage 0-2 isn't wasted capital — it's insurance against the tail-risk (severe) days that do the most damage.

Staffing cuts during outages may be a false economy

- Reduced staffing on high-stage days could cause lost sales beyond what power alone explains — e.g. being unable to serve a backlog of customers once power returns. Worth testing directly if hourly-level data becomes available.

Service businesses are structurally more exposed than product businesses

- This pattern is based on a single business per type in this dataset, so it should be generalised cautiously — but it is consistent with the general logic that services with no non-electric fallback have zero revenue during an outage, while product retailers can still transact.

4. Recommendations

- **1. Prioritise generator/inverter investment for Stage 3+ resilience —** especially for service-based businesses where "no power" means "no trade" at all. Even a small inverter for lighting and card machines could recover a meaningful share of the ~57% loss seen at the salon.
- **2. Avoid staffing cuts during high-stage days where possible —** pair scheduling decisions with a sales-per-staff measure to test whether maintained staffing offsets some of the severity-driven decline.
- **3. Plan cash-flow buffers around loadshedding schedules —** severe-stage days can approach zero revenue for the most exposed businesses; one business recorded 3 full zero-sales days in six weeks.
- **4. Shift trading hours or promotions to lower-stage windows —** concentrate marketing spend and promotions on days/times historically at Stage 0-2.
- **5. Investigate shared-resource models —** the source data's "borrowed neighbour generator" note suggests informal businesses are already doing this informally; formalising shared generator access among small traders could be cheaper than individual capital expenditure.

5. Caveats & Limitations

- Small sample: 4 businesses, ~33 days each. Suitable for demonstrating analysis technique, not for population-level claims about all South African small businesses.
- The generator effect is confounded with business type and size — only the retail store has a generator, so its advantage cannot be cleanly separated from other factors like scale, stock diversity, or footfall.
- Several rows required cleaning for data-quality issues (spelling inconsistencies, wrong province abbreviations, a repeated header row) before analysis — see Section 1 for the full audit trail.
- Analysis period (Jan-Feb 2023) is six weeks; seasonal effects (e.g. the Valentine's Day sales spike visible in the source notes) are not separated from the loadshedding effect.

End of report.